jetson查看imu数据

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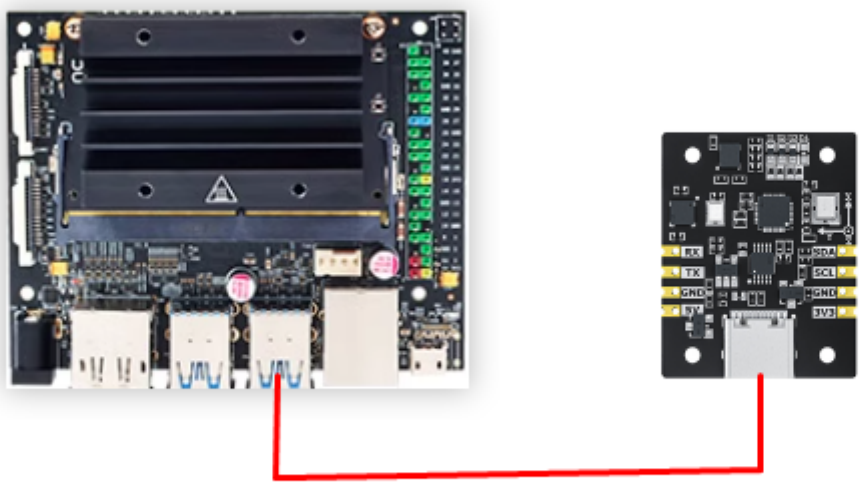
1.连接设备

本教程以jetson nano b01为例

将IMU姿态传感器通过type-c线插在主控的USB上。



2.查看设备状态

查看设备id

```
lsusb
```

```
jetson@yahboom: ~
jetson@yahboom:~$ lsusb
Bus 002 Device 002: ID 2109:0817 VIA Labs, Inc.
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 004: ID 0483:5750 STMicroelectronics
Bus 001 Device 003: ID 1a86:7523 QinHeng Electronics HL-340 USB-Serial adapter
Bus 001 Device 002: ID 2109:2817 VIA Labs, Inc.
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
jetson@yahboom:~$
```

查看设备号

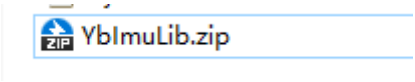
```
ll /dev/ttyU*
```

```
jetson@yahboom:~$ ll /dev/ttyU*
crwxrwxrwx 1 root dialout 188, 0 10月 20 19:38 /dev/ttyUSB0
jetson@yahboom:~$
```

3.安装驱动库

3.1下载python驱动库文件

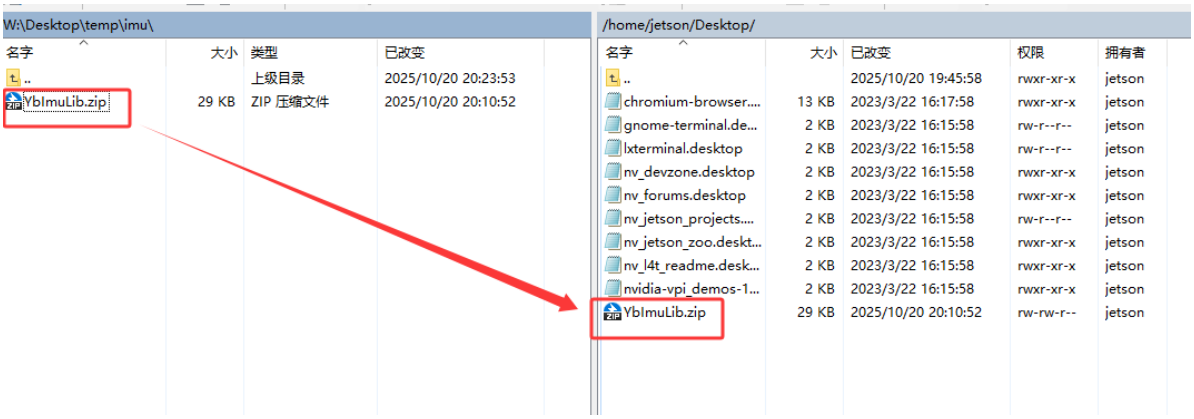
资料文件夹有提供最新版本的驱动库，名字为YbImuLib.zip



3.2传输文件

通过WinSCP软件将驱动库压缩包文件拖入Jetson桌面上。

安装成功后驱动库文件可删除掉。



如果还不会使用WinSCP传输文件的朋友，请查看以下网页WinSCP详细安装和操作方法：

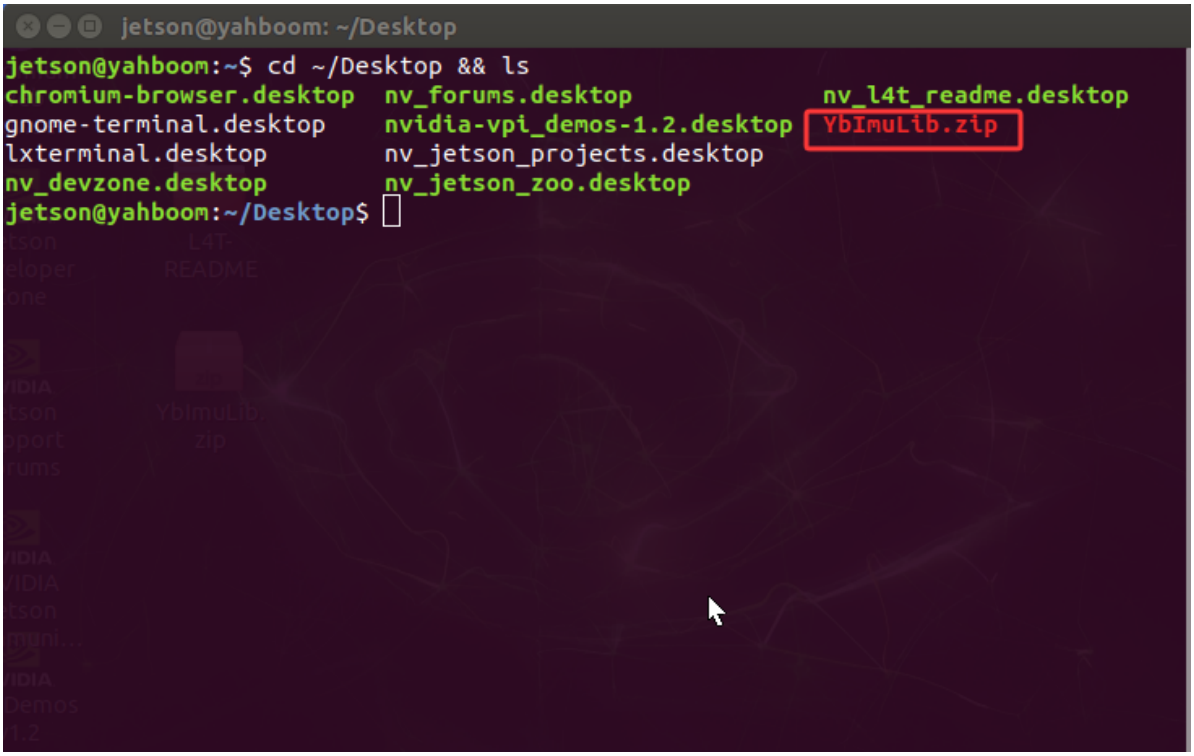
[文件传输 (Jetson Nano 4 GB)]

3.3安装驱动库

打开Jetson Nano的终端，输入以下命令解压。

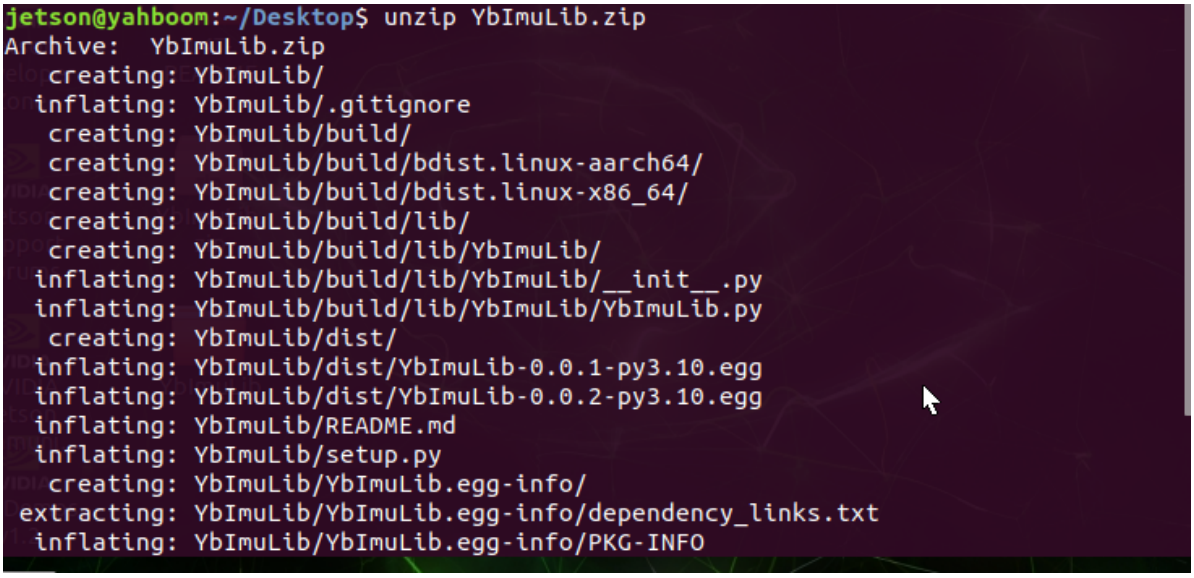
进入桌面并查看文件是否存在，红色方框中就是目标文件

```
cd ~/Desktop && ls
```



解压文件

```
unzip YbImuLib.zip
```



进入驱动库文件夹

```
cd YbImuLib
```

运行安装命令，看到最后提示的安装版本号则表示安装成功。此命令会覆盖掉之前已经安装过的Rosmaster_Lib驱动库。

```
sudo python3 setup.py install
```

```
jetson@yahboom:~/Desktop/YbImuLib$ sudo python3 setup.py install
[sudo] password for jetson:
running install
/usr/local/lib/python3.6/dist-packages/setuptools/command/install.py:37: SetuptoolsDeprecationWarning: setup.py install is deprecated. Use build and pip and other standards-based tools.
  setuptools.SetuptoolsDeprecationWarning,
/usr/local/lib/python3.6/dist-packages/setuptools/command/easy_install.py:159: EasyInstallDeprecationWarning: easy_install command is deprecated. Use build and pip and other standards-based tools.
  EasyInstallDeprecationWarning,
/usr/local/lib/python3.6/dist-packages/pkg_resources/__init__.py:119: PkgResourc
```

安装需要的库

```
sudo pip3 install pyserial
sudo pip3 install smbus2
```

4.查看imu数据

参考**3.2传输文件**的操作，用winscp将**YbImu_ReadData_Serial.py**文件传输到jetson上，然后运行命令

```
python3 YbImu_ReadData_Serial.py
```

```
jetson@yahboom:~/Desktop$ python3 YbImu_ReadData_Serial.py
YbImu Serial Opened! Baudrate=115200
----------------create receive threading--------------
Firmware version: V0.0.9
Press Ctrl+C to exit the program.
------ Sensor Data ------
Acceleration [g]:       x=-0.230, y=-0.013, z= 0.980
Gyroscope [rad/s]:      x= 0.000, y= 0.000, z= 0.000
Magnetometer [uT]:      x= 18.262, y= 9.107, z=-13.404
Quaternion:             w=-0.95560, x=-0.02311, y=-0.11265, z= 0.26501
Euler Angle [deg]:      roll=-0.91, pitch= 13.15, yaw=-31.00
Barometer:              height= 0.00 m, temperature= 0.00 °C
                        pressure= 0.00000 Pa, pressure_diff= 0.00000 Pa
--------------------------
------ Sensor Data ------
Acceleration [g]:       x=-0.229, y=-0.018, z= 0.978
Gyroscope [rad/s]:      x=-0.012, y= 0.000, z= 0.000
Magnetometer [uT]:      x= 18.165, y= 8.985, z=-13.477
Quaternion:             w=-0.95546, x=-0.02221, y=-0.11261, z= 0.26560
Euler Angle [deg]:      roll=-1.02, pitch= 13.12, yaw=-31.09
Barometer:              height= 0.00 m, temperature= 0.00 °C
                        pressure= 0.00000 Pa, pressure_diff= 0.00000 Pa
```

注意：以上为9轴IMU的数据读取，6轴无磁力计（Magnetometer）与气压计（Barometer）数据，9轴无气压计（Barometer）数据。

5.注意事项

如果orin系列主板可以查到设备ID，但是无法查到设备号，可以参考下面的命令安装ch34x驱动

```
sudo apt remove brltty
git clone https://github.com/clhchan/CH341SER.git
cd CH341SER
make -j6
sudo make install
sudo modprobe ch34x
```